

Chapter 11 — Perioperative Care in Urologic Surgery

- **Perioperative care keeps evolving** with new tools and an ageing population; economics drive more outpatient surgery, shorter stays and fewer complications.
- **Standard of care:** even complex urological operations are admitted on the day of surgery and discharged the day after.
- **Despite the shift to office-based and non-surgical care**, the diversity of genitourinary disease requires every urologist to command perioperative principles.

PREOPERATIVE EVALUATION

- **Patient acuity is rising** with age and comorbidity.
- **Surgeon and patient share responsibility** for evaluation by the wider team, so the patient arrives in, and leaves, theatre optimised.
- **Appropriate specialist consultation** improves outcomes and reduces cancellations.

PRESURGICAL TESTING

- **Goal:** find an undiagnosed comorbidity, an undertreated condition, or a significant exacerbation that may affect the operative course.
- **Individualise** by age, history, examination and procedure.
- **Routine testing beyond history and examination has never been shown cost-effective**, and predicts morbidity **less well than American Society of Anesthesiologists (ASA) status** or American Heart Association/American College of Cardiology (AHA/ACC) risk guidance; a systematic review found no support for routine testing before elective non-cardiac surgery.
- **Typical panel:** full blood count; basic metabolic panel; prothrombin time (PT), partial thromboplastin time (PTT), international normalised ratio (INR) (controversial); electrocardiogram (ECG); ABO/Rh group and antibody screen; chest radiograph.

COMPARISON: WHEN EACH PREOPERATIVE TEST IS INDICATED

Test	Indication
PT/PTT	Omit in most; controversial unless on anticoagulation or with personal/surgical bleeding history
Urine pregnancy test	Every woman of childbearing age on the morning of surgery , unless ovaries or uterus removed
ECG	Age >40 years or any cardiac history (predictive value debated — studies conflict)
Chest radiograph	Not routine without cardiopulmonary disease

- **ASA Task Force on Pre-anaesthesia Evaluation** would say only that tests "may be ordered ... on a selective basis" to guide management.

American Society of Anesthesiologists Classification and Risk

- **About 27 million US operations a year; about 8 million (30%) in patients with significant coronary or other cardiac disease** — hence the cardiovascular focus.
- **ASA classification** (1961, revised to six classes; Box 11.1) grades **overall physical status, not surgical risk**; subjective yet an **independent predictor of mortality**.
- **Multivariate scores:** Goldman's criteria (Table 11.1) sum weighted points; the **Modified Cardiac Risk Index** (Table 11.2) simplifies to **six predictors**.

BOX 11.1 AMERICAN SOCIETY OF ANESTHESIOLOGISTS (ASA) CLASSIFICATION

Class	Definition
I	Normal healthy patient
II	Mild systemic disease
III	Severe systemic disease that limits activity but is not incapacitating
IV	Incapacitating disease that is a constant threat to life
V	Moribund , not expected to survive 24 h with or without operation
VI	Declared brain-dead organ donor
E	Suffix for emergency surgery, added to classes I–V

- **Trap:** III vs IV turns on **limits activity** vs **constant threat to life**; "E" is a suffix, not a class, and never attaches to VI.

COMPARISON: TABLE 11.1 GOLDMAN'S CARDIAC RISK INDEX AND TABLE 11.2 MODIFIED CARDIAC RISK INDEX (BOTH MODIFIED FROM AKHTAR AND SILVERMAN, 2004)

Goldman — risk factor (points)	Modified Cardiac Risk Index — predictor (1 point each)
Third heart sound or jugular venous distension (11)	Ischaemic heart disease
Recent myocardial infarction (10)	Congestive heart failure
Non-sinus rhythm or premature atrial contractions on ECG (7)	Cerebrovascular disease
>5 premature ventricular contractions (7)	High-risk surgery
Age >70 years (5)	Preoperative insulin for diabetes
Emergency operation (4)	Preoperative creatinine >2 mg/dL
Poor general medical condition (3); intrathoracic, intraperitoneal or aortic surgery (3); significant aortic stenosis (3)	—
Risk of cardiac complication: 6–12 points → 7%; 13–25 → 14%; ≥26 → 78%	Each additional predictor increases perioperative cardiovascular risk

- **What the pair tests:** Goldman weights nine factors, heaviest for **active heart failure (S3/JVD)** and **recent MI**; the modified index counts six equally weighted predictors.
- **Pattern:** both include heart failure and surgery type; only the modified index includes **insulin-treated diabetes, cerebrovascular disease and creatinine**; only Goldman includes **age, rhythm and aortic stenosis**.
- **Trap:** the modified index is an **unweighted count** — do not carry Goldman's weights into it.

PREOPERATIVE CARDIOVASCULAR EVALUATION

- **Initial history, examination and ECG** seek coronary disease, heart failure, symptomatic arrhythmia, pacemaker/implantable defibrillator, and orthostatic hypotension; define **severity and stability**.
- **Cardiac risk is modified** by functional capacity, age, diabetes, peripheral vascular disease, renal dysfunction and chronic obstructive pulmonary disease (COPD).
- **ACC/AHA guidelines use three categories:** clinical markers, functional capacity, and type of surgery.

Clinical Markers

COMPARISON: CLINICAL PREDICTORS OF PERIOPERATIVE CARDIOVASCULAR RISK

Major	Intermediate	Minor
Acute myocardial infarction (MI) <7 days; recent MI 7 days–1 month; unstable angina; any ischaemic burden (symptoms or non-invasive test); decompensated heart failure; significant arrhythmia; severe valvular disease	Mild angina; prior MI by history or Q waves; compensated heart failure; diabetes; renal insufficiency (creatinine >2 mg/dL)	Advanced age; abnormal ECG; non-sinus rhythm (e.g. atrial fibrillation); prior stroke; uncontrolled hypertension

- **Examinable contrasts:** decompensated (major) vs compensated (intermediate) heart failure; recent (major) vs prior (intermediate) MI.
- **Timing after MI:** the old **3–6-month** wait is abandoned; ACC stratifies by infarct severity and reinfarction risk on stress testing; without trial data, **waiting 4–6 weeks before elective surgery is reasonable**, considering stents/bypass and antiplatelet needs.

Functional Capacity

- **Measured in metabolic equivalents (METs); 4 METs ≈ climbing two flights of stairs** — cheap and simple.
- **Duke Activity Status Index (DASI)** (Table 11.3), validated repeatedly, quantifies capacity.
- **≥4 METs → no further invasive cardiac evaluation needed.**

TABLE 11.3 DUKE ACTIVITY STATUS INDEX (MODIFIED FROM HLATKY ET AL., 1989)

Activity ("Can you ...?")	Yes
Take care of yourself (eating, dressing, bathing, toileting)	2.75
Walk indoors, around the house	1.75
Walk a block or two on level ground	2.75
Light housework (dusting, washing dishes)	2.70
Moderate housework (vacuuming, sweeping, carrying groceries)	3.50
Yard work (raking, weeding, pushing a power mower)	4.50
Have sexual relations	5.25
Climb a flight of stairs or walk up a hill	5.50
Moderate recreation (golf, bowling, dancing, doubles tennis, throwing a ball)	6.00
Strenuous sport (swimming, singles tennis, football, basketball, skiing)	7.50
Run a short distance; heavy housework (scrubbing floors, moving furniture)	8.00 each

- **Scoring:** "No" = 0; DASI = sum of all 12 items. **Peak VO₂ (mL/kg/min) = 0.43 × DASI + 9.6; METs = peak VO₂ ÷ 3.5 (× 0.286).**
- **Pattern:** items rise in weight with aerobic demand; running and heavy housework score highest (8.00).

Surgery-Specific Cardiac Risk

- **Determined by type of surgery and degree of haemodynamic stress.**

Risk	Procedures
High	Major emergency surgery (especially elderly); long operations with large fluid shifts or blood loss
Intermediate	Intraperitoneal, laparoscopic and robotic-assisted laparoscopic surgery

Risk	Procedures
Low	Endoscopic and superficial (no body-cavity entry) procedures

Concept — Cardiac risk is the product of three independent axes: how sick the heart is (clinical markers), how much reserve the patient has (METs), and how much stress the operation imposes (surgery type). A patient with good reserve undergoing low-stress surgery needs no further testing whatever the history.

PULMONARY EVALUATION

- **Most important for thoracic and abdominal surgery.**
- **Pulmonary function testing** (forced expiratory volume in 1 second (FEV₁), forced vital capacity, diffusing capacity for carbon monoxide) for known lung disease, significant smoking history or symptoms — gives a baseline.
- **FEV₁ <0.8 L or <30% predicted → high risk.**
- **Specific risk factors:** COPD, smoking, sputum production, pneumonia, dyspnoea, obstructive sleep apnoea.
- **Smokers:** 4-fold postoperative pulmonary morbidity and up to 10-fold mortality.
- **Restrictive fares better than obstructive disease** — preserved maximal expiratory flow allows an effective cough with less sputum.
- **General factors:** age, low albumin, obesity, impaired sensorium, prior stroke, immobility, acute renal failure, chronic steroids.

HEPATOBIILIARY EVALUATION

- **Advanced liver disease survival has improved**, so these patients come to surgery more often; mild-moderate disease is often asymptomatic and must be sought.
- **History:** prior hepatitis, timing and precipitants (important if staff have a blood-borne exposure); pruritus, bleeding, distension, weight gain.
- **Examination:** jaundice/scleral icterus visible at bilirubin >3 mg/dL; caput medusae, palmar erythema, spider naevi, clubbing; severe — ascites, encephalopathy, asterixis, cachexia.

COMPARISON: CHILD CLASSIFICATION VS MODEL FOR END-STAGE LIVER DISEASE (MELD)

	Child classification	MELD score
Inputs	Bilirubin, albumin, PT + clinical encephalopathy and ascites	Bilirubin, creatinine, INR (linear regression); calculator adds age and ASA class
Nature	Partly subjective; arbitrary cut-offs	Objective, weighted, no arbitrary cut-offs → more accurate
Output	Surgical mortality: A 10%, B 30%, C 76–82%; also predicts liver failure, encephalopathy, bleeding, infection, renal failure, hypoxia, ascites	7-, 30-, 90-day, 1- and 5-year surgical mortality; tracks 30-day mortality even in colorectal surgery without liver disease

- **Other independent mortality risks:** emergency surgery, COPD.
- **The two scores are complementary** in cirrhotic patients.

SPECIAL POPULATIONS

Elderly

- **By 2050 Americans >65 will more than double to 89 million**, >20% of them over 85; octogenarian and nonagenarian surgery is increasing.
- **Whether age or comorbidity drives risk is unsettled**, but **age independently predicts morbidity and mortality**; after radical cystectomy, 90-day mortality climbs across the <70, 70–80 and >80 bands.
- **Over 80**: hypertension and dyspnoea are the commonest comorbid risks; preoperative transfusion, emergency surgery and weight loss best predict morbidity; **longer operative time raises mortality** in octogenarians.
- **Delirium** — often dismissed as "sundowning" — may be the **first sign of metabolic or infective complication**; assess cognition, function and mood preoperatively.
- **Reduce delirium**: minimise noise, familiar surroundings, consistent sleep, **regional anaesthesia** where possible, **minimal opioids**.

Morbid Obesity

- **Weigh operative risk against natural history** when timing surgery; **BMI ≥ 45 kg/m² costs 8–13 years** of life expectancy.
- **Cardiac symptoms are non-specific** and examination underestimates dysfunction; **>3 coronary risk factors → consider non-invasive cardiac evaluation**.
- **Comorbidities**: ASCVD, heart failure, hypertension, sleep-apnoea pulmonary hypertension, hypoventilation, arrhythmias, deep vein thrombosis/pulmonary embolism, poor exercise capacity; ventilation–perfusion mismatch.
- **Positioning injury and wound infection risk** → consider laparoscopy when appropriate.

Pregnancy

- **Commonest reason**: renal colic/stones, found on fetal ultrasound or during colic.
- **Fetal radiation risk highest from preimplantation to ~15 weeks**; **malformation threshold 10 cGy** → ultrasound first (CT abdomen ~1 cGy; intravenous pyelogram ~0.3 cGy).
- **Delay surgery until the fetus is mature** unless mother or fetus would be harmed.

Trimester	Anaesthetic/surgical risk
First	Teratogenic exposure to anaesthetic agents; delay semi-elective surgery beyond it
Second	Safest — organogenesis complete; almost no risk of anaesthetic malformation or miscarriage
Third	Preterm labour; fetal hypoxaemia from altered uterine flow and maternal acid–base change

- **Involve obstetrician, perinatologist and anaesthetist**; fetal heart rate and tocometry before and after.
- **Analgesia: opioids acceptable** (no human birth defects at normal doses; wean early — fetal dependency); avoid NSAIDs — **premature closure of the ductus arteriosus**.

Nutritional Status

- **Malnutrition impairs host defence**, wound healing, immunity, bowel recovery and organ function.
- **Assessment**: weight loss plus lymphocyte count and albumin (visceral protein); **>20 lb (~9 kg) loss in 3 months = severe malnutrition**; validated tools include the Subjective Global Assessment.

COMPARISON: PARENTERAL VS ENTERAL NUTRITION

	Total parenteral nutrition (TPN)	Enteral nutrition
Use	Severe malnutrition with a non-functioning gut	Preferred whenever feasible (nasogastric tube, gastrostomy, jejunostomy)
Evidence	7–10 days preoperatively improves outcome if undernourished; no benefit or more sepsis if well or mildly undernourished	5–20 days preoperatively cuts complications by 10–15%
Physiology	—	Maintains gut-associated lymphoid tissue, mucosal blood flow and barrier; formulas 1–2 kcal/mL, lactose-free, full vitamins/minerals in <2 L/day
Postoperative	If requirements cannot be met within 7–10 days ; not useful routinely if well-nourished or eating within a week	Monitor nutrition, gastrointestinal tolerance, fluid/electrolytes
Complications	Line insertion: pneumo/haemothorax, chylothorax (thoracic duct); line sepsis — commonest, needs removal ; venous and catheter thrombosis (thrombolytics)	Tube dislodgement → pulmonary/peritoneal complications; adynamic ileus

Concept — *Each special population reshapes the same risk calculation: the elderly lose reserve and present atypically (delirium), the obese mask cardiac disease, pregnancy adds a second patient with a time-dependent vulnerability, and malnutrition removes the capacity to heal — so the gut should be used whenever it works.*

PREPARATION FOR SURGERY**Preoperative Optimisation Strategies****Optimisation of Comorbid Illness***Cardiac.*

- **Of nitrates, calcium-channel blockers and β -blockers, only β -blockade improved outcome.**
- **Mangano (NEJM, 1996)**, landmark: prophylactic **atenolol** improved outcomes after non-cardiac surgery; a very large retrospective study linked perioperative β -blockade to lower mortality in **high-risk** major non-cardiac surgery.
- **Goal-directed therapy** (judicious fluids, inotropes, oxygen to physiological targets) further reduces risk; an early trial showed a large fall in mortality.

Pulmonary.

- **Stop smoking ≥ 8 weeks before surgery.** Stopping <8 weeks may **raise** risk — loss of smoke irritation reduces coughing and pulmonary toilet. **>6 months' abstinence → risk equals non-smokers.**
- **COPD:** preoperative bronchodilators dramatically reduce complications; treat infection with antibiotics; **steroid pretreatment for asthma**; epidural/regional anaesthesia, physiotherapy and continued bronchodilation help.

Diabetes.

- **Hyperglycaemia** impairs healing and raises infection; **hypoglycaemia under anaesthesia may go unrecognised.**
- **Stop long-acting oral hypoglycaemics;** prefer short-acting agents or sliding-scale insulin; **target glucose 80–250 mg/dL;** frequent fingerstick checks postoperatively; resume usual regimen when eating.

- **Insulin pump: continue basal infusion on the day of surgery**, correct with the pump; know the patient's sensitivity factor.

Thyroid.

- **Either dysfunction → endocrinologist; defer elective surgery until euthyroid.**

COMPARISON: HYPERTHYROID VS HYPOTHYROID PATIENT

	Hyperthyroid	Hypothyroid
Greatest risk	Thyroid storm: fever, tachycardia, confusion, cardiovascular collapse; atrial fibrillation in ~20% ; airway compression/deviation by goitre	Increased sensitivity to anaesthetics and opioids; severe — myocardial dysfunction, coagulopathy, electrolyte disturbance, reduced gut motility
Features	—	Lethargy, cold intolerance, hoarseness, constipation, dry skin, apathy, periorbital oedema, thin eyebrows, brittle hair, hypothermia, bradycardia, slow-relaxing reflexes
Management	Continue propylthiouracil/methimazole and β-blocker on the day of surgery; storm → iodine and steroids	Confirm (low thyroxine, high thyroid-stimulating hormone) → levothyroxine

Corticosteroids and the hypothalamic–pituitary–adrenal (HPA) axis.

Exposure	HPA status and action
≤5 mg prednisone/day, any duration	No clinically significant suppression; no supplement — give the usual dose
>20 mg prednisone/day for >3 weeks, or clinically cushingoid	Presume suppressed
Potent topical steroid ~2 g/day; inhaled ~0.8 mg/day	Can suppress
Suppressive dose within 1 year of surgery	Supplement; low-dose adrenocorticotrophic hormone (ACTH) stimulation test can assess

- **Stress dosing: hydrocortisone 50–100 mg intravenously (IV) before induction → 25–50 mg every 8 h for 24–48 h** until usual dose resumes.
- **Minor procedures under local anaesthesia need no stress dose.**

Management of Medications — Antithrombotic Therapy

- **Warfarin and antiplatelets cause significant postoperative bleeding** (unlike venous thromboembolism (VTE) prophylaxis) → balance bleeding against thrombosis.
- **Warfarin** (atrial fibrillation, mechanical valves, prior VTE): **half-life 36–42 h → stop 5 days before surgery for INR <1.5.**
- **Direct oral anticoagulants** (apixaban, dabigatran, rivaroxaban) differ pharmacologically; the surgeon must know each.
- **Bridging** depends on thrombotic risk (Table 11.4): **2012 American College of Chest Physicians (ACCP) — bridge moderate- and high-risk patients** with therapeutic subcutaneous low-molecular-weight heparin (LMWH) or IV unfractionated heparin (UFH).

TABLE 11.4 RISK STRATIFICATION FOR ARTERIAL OR VENOUS THROMBOEMBOLISM EVENTS DURING PERIOPERATIVE PERIOD IN PATIENTS ON CHRONIC ANTICOAGULANT THERAPY (MODIFIED FROM DOUKETIS ET AL., ACCP, 2012)

Risk	Mechanical heart valve	Atrial fibrillation (CHADS ₂)	Venous thromboembolism
Low	Bileaflet aortic valve, no AF, no stroke risk factors	0–2, no prior stroke/TIA	Single VTE >12 months ago, no other risk
Moderate	Bileaflet aortic valve + ≥1 of AF, prior stroke/TIA, hypertension, diabetes, heart failure, age >75	3–4	VTE within 3–12 months; non-severe thrombophilia (heterozygous factor V Leiden or factor II); recurrent VTE; active cancer (treated ≤6 months or palliative)
High	Any mitral prosthesis; caged-ball or tilting-disc aortic; stroke/TIA within 6 months	5–6; stroke/TIA within 3 months; rheumatic valve disease	VTE within 3 months; severe thrombophilia (protein C, S or antithrombin deficiency, antiphospholipid antibodies, multiple abnormalities)

- **CHADS₂** = congestive heart failure, hypertension, age, diabetes, stroke; TIA = transient ischaemic attack.
- **Pattern:** recency of event, valve position/type and severity of thrombophilia define the high stratum. **Trap:** any **mitral** prosthesis is high risk; a bileaflet **aortic** valve is low unless other factors.

Antiplatelet therapy and coronary stents.

- Cardiovascular prevention use is uncontroversial; stent thrombosis prevention is the dilemma.
- Aspirin and clopidogrel are irreversible → stop 7–10 days before surgery to minimise bleeding.
- Premature interruption → 25–50% risk of significant MI, with raised mortality.

	Bare-metal stent	Drug-eluting stent
Dual antiplatelet duration	6 weeks	12 months
Delay elective urological surgery	≥30 days	>1 year if possible, then keep at least single-agent therapy (late thrombosis described)
Surgery cannot wait (e.g. cancer)	ACCP: continue aspirin and clopidogrel perioperatively (drug-eluting stents); close urologist–cardiologist communication	

- **Two bare-metal figures, two sources:** 6 weeks is the recommended dual-therapy duration; ≥30 days is the minimum surgical delay from a literature review.

Bowel Preparation

- **Rationale:** fewer faeces and bacteria → less anastomotic leak, abscess, wound infection.
- **Flora:** aerobes — *Escherichia coli*, *Enterococcus faecalis*; anaerobes — *Bacteroides*, *Clostridium*.
Concentration (organisms/g): jejunum 10–10⁵; distal ileum 10⁵–10⁷; ascending colon 10⁶–10⁸; **descending colon 10¹⁰–10¹².**
- Evidence is inferred from colorectal surgery.

COMPARISON: ORAL ANTIBIOTIC VS MECHANICAL BOWEL PREPARATION

	Oral antibiotic preparation	Mechanical bowel preparation (MBP)
Standard	Neomycin + erythromycin (Nichols–Condon trial, 1977: septic complications 43% MBP alone vs 9% MBP + oral antibiotics)	Formerly days of laxatives, lavage, enemas; now polyethylene glycol or sodium phosphate, as outpatient

	Oral antibiotic preparation	Mechanical bowel preparation (MBP)
Current evidence	Benefit debated since routine IV prophylaxis; Cochrane: aerobic + anaerobic cover (oral, IV or both) cuts wound infection up to 75%; unknown if useful when colon not emptied	No benefit for leak, abscess or mortality (meta-analysis of randomised controlled trials (RCTs)); overall surgical site infection (SSI) higher with MBP; Cochrane agrees
Downside	<i>Clostridioides difficile</i> colitis	Discomfort, electrolyte disturbance
Practice	Still used by most colorectal surgeons	Near-universal belief; urological series show equivalent SSI after cystectomy/diversion without it

- **Polyethylene glycol preferred** in the elderly and in renal insufficiency, heart failure, electrolyte disturbance and cirrhosis — **completely non-absorbable**.
- **Exceptions where preparation persists: enema before transrectal prostate biopsy** (portal of entry, bacteraemia); laparoscopy (believed to improve exposure and ileus — **unproven**).
- **Postoperative ileus**: some primary ileus is normal; **>72–96 h → evaluate for mechanical obstruction** (adhesions, intra-abdominal process, retroperitoneal haemorrhage).
- **Reduce ileus**: minimise opioids, selective nasogastric tubes, correct electrolytes; **gum chewing (sham feeding)** speeds recovery; **alvimopan** (peripheral opioid antagonist, FDA 2008) shortened stay after radical cystectomy by over 2 days in a phase IV trial. Both feature in enhanced recovery after surgery (ERAS) pathways.

Concept — *Optimisation means reversing what can be reversed before the insult of surgery, but the timing rules are counterintuitive: late smoking cessation, premature antiplatelet withdrawal and routine mechanical bowel preparation can each do more harm than good.*

INTRAOPERATIVE MANAGEMENT

Patient Environment

- **Hypothermia is harmful in elective surgery** (therapeutic only in some trauma/brain injury).
- **Mechanism**: induction → peripheral vasodilatation redistributes core heat (early drop); thereafter **radiation and conduction** account for most loss.
- **Normothermia = core 36–38°C**; even **1–2°C** of hypothermia is harmful — mild hypothermia increases blood loss and transfusion (meta-analysis of RCTs) via impaired clotting enzymes and platelet aggregation.
- **SSI risk rises with mild hypothermia (34–36°C)** — impaired immunity and vasoconstriction-driven tissue hypoxia.
- **Kurz (1996)**, landmark, 200 colorectal patients: hypothermia → **3× wound infection and +2.6 days' stay**. The **Surgical Care Improvement Project (SCIP)** includes normothermia.
- **Warming**: blankets, warmed IV fluids, **warmed irrigation (transurethral resection of the prostate (TURP), long endoscopy)**, warmed humidified CO₂ in laparoscopy, higher theatre temperature.

Patient Safety

- **Harvard Medical Practice Study (Brennan, 1991)**: 48% of adverse events accompanied an operation → inspired the Institute of Medicine's *To Err Is Human*. The **operating theatre is the commonest venue** of preventable surgical injury; the surgeon is "captain of the ship" but every member must attend to detail.
- **Three immediately preventable injuries**: retractor, thermal and positioning.
- **Self-retaining retractors → femoral neuropathy**: lateral blades must cradle rectus only, **never compress psoas**; reinspect periodically.

- **Thermal:** diathermy, argon beam, bipolar, lasers; **light-cable ends burn through drapes** → off/standby when not in use.
- **Morbid obesity:** hydraulic table, long instruments, extra padding, wide compression sleeves, table extensions.

Patient Positioning

- **Shared team responsibility;** knowledge drawn from anaesthesia. The ASA Closed Claims Database (1999) found **neuropathy the second-leading cause of liability** → ASA practice advisory (Box 11.2).
- **Mechanism of position-related neuropathy:** stretch, prolonged compression or ischaemia.

COMPARISON: UROLOGICAL POSITIONS AND THEIR HAZARDS

Position	Key risks and rules
Supine (safest)	Abduction >90° → brachial plexus; pad arm boards (ulnar groove; spiral groove → radial nerve); tucked arms — avoid hand/forearm pressure, detect IV infiltration early (forearm compartment syndrome)
Lithotomy	Persistent neuropathy rare, but transient neuropathy occurs in a small minority, almost all resolving by 6 months. Move both legs together; hips flexed 80–100°, abducted 30–45°; pad legs off stirrups; guard fingers in stirrup joints; adapt for contractures
Lateral decubitus / flank	Anterior and posterior support; ipsilateral arm on chest/rest/pillows, <90° abduction; contralateral arm on padded board; axillary roll just caudal to (not in) the axilla; tilt the table fully before draping to test security
Prone (percutaneous nephrolithotomy, PCNL)	Pad torso, elbows, hips, legs; secure tube and lines; coordinated transfer; trolley immediately available for airway rescue
Steep Trendelenburg (robotic pelvis)	Respiratory and cardiovascular change, raised central venous and intracranial pressure; secure against cephalad slide; avoid fixed shoulder braces (brachial plexus)

BOX 11.2 AMERICAN SOCIETY OF ANESTHESIOLOGISTS TASK FORCE RECOMMENDATIONS ON THE PREVENTION OF PERIOPERATIVE PERIPHERAL NEUROPATHIES (ANESTHESIOLOGY, 2000)

Domain	Recommendation
Preoperative	Check the patient tolerates the planned position
Upper limb	Abduction ≤90° supine (prone tolerates more); avoid ulnar (postcondylar) groove pressure; tucked arm → neutral forearm; on arm board → supination or neutral; avoid spiral-groove (radial) pressure; elbow hyperextension stretches median nerve
Lower limb	Hamstring stretch in lithotomy → sciatic; peroneal nerve at fibular head; hip flexion/extension does not raise femoral neuropathy risk
Padding	Padded arm boards; chest rolls in lateral position; elbow and fibular-head padding
Equipment	Functioning automated blood-pressure cuffs do not add risk; shoulder braces in steep head-down may cause brachial plexopathy
Postoperative / documentation	Simple nerve check aids early recognition; charting positioning focuses attention and feeds quality improvement

- **Trap:** femoral neuropathy comes from **retractor compression of psoas**, not hip position.

Antibiotic Prophylaxis

- **CDC (1999):** SSIs are **~40% of nosocomial infections** in surgical patients and add **7–10 days** in hospital; nationally ~1 million extra inpatient days. Level 1 evidence supports prophylaxis across procedures; Medicare withholds payment for some SSI admissions → prevention is mandatory.

- **Adjuncts:** hand-washing, scrub, sterile field; hair removal does not reduce SSI but if done use **clippers or depilatory cream, not razors**.
- **Three determinants of need:** (1) patient susceptibility (Box 11.3); (2) procedural risk — wound class (Box 11.4); (3) consequence of infection (e.g. cystoscopy for microhaematuria: no prophylaxis in a young healthy patient, yes in an elderly insulin-dependent diabetic).

BOX 11.3 PATIENT FACTORS THAT INCREASE THE RISK OF INFECTION

- Advanced age; anatomical anomalies; poor nutrition; smoking; chronic corticosteroids; immunodeficiency; chronic indwelling hardware; infected endogenous/exogenous material; distant coexistent infection; prolonged hospitalisation. **They act by reducing defences, raising local bacterial load or altering flora.**

BOX 11.4 SURGICAL WOUND CLASSIFICATION (DATA FROM CRUSE, 1992; MANGRAM ET AL., 1999; GARNER, 1985)

Class	Definition
Clean	Uninfected, no inflammation, no entry into genital, urinary or alimentary tract; primary closure, closed drainage
Clean-contaminated	Uninfected, controlled entry into these tracts; primary closure, closed drainage
Contaminated	Major break in sterile technique (gross gastrointestinal spillage, non-purulent inflammation); fresh accidental wounds
Dirty-infected	Pre-existing clinical infection or perforated viscus; old traumatic wounds with devitalised tissue

- **Trap:** most urological operations entering the urinary tract are **clean-contaminated**; transrectal biopsy is **contaminated**.
- **Keys once prophylaxis is chosen: the right agent, correct timing, correct dose.**
- **Agent:** procedure dictates pathogens (TURP → urinary organisms; sigmoid diversion → anaerobes); local resistance matters — level 1 evidence supports fluoroquinolones for endoscopy but ***E. coli* resistance** is changing practice; use the monthly **hospital antibiogram**. AUA amendment (2012) for prostate biopsy: **cephalosporins and/or aminoglycosides** in resistant communities.
- **Timing:** **Classen (1992)**, pivotal — within 2 h of incision; SCIP — **within 60 min**; a multicentre trial favoured **≤30 min over 31–60 min** and confirmed that giving it **at or after incision roughly doubles SSI**.
- **Duration:** **≤24 h** without established infection (longer → *C. difficile*, resistance, cost).
- **Dose:** by weight, renal and hepatic function; **redose if the case exceeds 4 h**. Compliance with AUA best practice remains **<60%**.

TABLE 11.5 AMERICAN UROLOGICAL ASSOCIATION BEST PRACTICE STATEMENT ON RECOMMENDED ANTIMICROBIAL PROPHYLAXIS FOR UROLOGIC PROCEDURES (MODIFIED FROM LIGHTNER ET AL., 2020)

Procedure (wound class)	Organisms	Indicated	First choice	Alternative	Duration
Cystourethroscopy ± minor manipulation, biopsy, fulguration (CC)	GNR, rarely enterococci	Uncertain — host factors; not needed if urine sterile	TMP-SMX; amoxicillin-clavulanate	1st/2nd-gen cephalosporin; AG (aztreonam) ± ampicillin	Single
Transurethral (TURP, TURBT, laser enucleation/ablation) (CC)	GNR, rarely enterococci	All	Cefazolin; TMP-SMX	Amox-clav; AG ± ampicillin	Single
Brachytherapy, cryotherapy (CC)	<i>S. aureus</i> , skin; GNR	All	Cefazolin	Clindamycin	Single

Procedure (wound class)	Organisms	Indicated	First choice	Alternative	Duration
Transrectal prostate biopsy (contaminated)	GNR, anaerobes; consider MDR if antibiotics ≤ 6 months, travel, health worker	All	Fluoroquinolone; 1st–3rd-gen cephalosporin (ceftriaxone) + AG	Aztreonam; consider ID advice	Single
Percutaneous renal surgery (CC)	GNR, enterococci, skin	All	1st/2nd-gen cephalosporin; AG + metronidazole or clindamycin	Ampicillin-sulbactam	≤ 24 h
Ureteroscopy (CC)	GNR, rarely enterococci	All; uncertain for simple diagnostic	TMP-SMX; 1st/2nd-gen cephalosporin	AG $\pm$ ampicillin; amox-clav	Single
Open/lap/robotic, urinary tract not entered (clean)	<i>S. aureus</i> , skin	Consider; may not be required	Cefazolin	Clindamycin	Single
Penile (circumcision, biopsy)	<i>S. aureus</i>	Likely not required	—	—	—
Urethroplasty	GNR, enterococci, <i>S. aureus</i>	Likely required	Cefazolin	Cefoxitin, cefotetan, ampicillin-sulbactam	Single
Controlled urinary-tract entry (nephrectomy, pyeloplasty, prostatectomy, partial cystectomy) (CC)	GNR (<i>E. coli</i>)	All	Cefazolin; TMP-SMX	Ampicillin-sulbactam; AG + metronidazole or clindamycin	Single
Small bowel (conduit, diversion) (CC)	Skin, <i>S. aureus</i> , GNR	All	Cefazolin	Clindamycin + AG; cefuroxime; aminopenicillin/ β -lactamase inhibitor + metronidazole	Single
Large bowel (colon conduit) (CC)	GNR, anaerobes	All	Cefazolin + metronidazole; cefoxitin; cefotetan; ceftriaxone + metronidazole; ertapenem	Ampicillin-sulbactam; ticarcillin-clavulanate; piperacillin-tazobactam	Single parenteral
Prosthetics (AUS, IPP, sacral neuromodulator) (clean)	GNR, <i>S. aureus</i> ; rising anaerobes, fungi	All	AG + 1st/2nd-gen cephalosporin or vancomycin	Aminopenicillin/ β -lactamase inhibitor	≤ 24 h
Inguinal/scrotal (orchidectomy, vasectomy, varicocele, hydrocele) (clean)	GNR, <i>S. aureus</i>	All (increased risk)	Cefazolin	Ampicillin-sulbactam	Single

Procedure (wound class)	Organisms	Indicated	First choice	Alternative	Duration
Vaginal/female incontinence (sling, fistula, diverticulum) (CC)	<i>S. aureus</i> , streptococci, enterococci, vaginal anaerobes	All	2nd-gen cephalosporin (better anaerobic cover); cefazolin equivalent for slings	Ampicillin-sulbactam; AG + metronidazole or clindamycin	Single
Shock-wave lithotripsy (clean)	GNR, enterococci	Only with risk factors	TMP-SMX; cefazolin; cefuroxime	Amox-clav; ampicillin + AG; clindamycin	Single

- **Abbreviations:** CC, clean-contaminated; TURBT, transurethral resection of bladder tumour; GNR, Gram-negative rods (*E. coli*, *Proteus*, *Klebsiella*; plus *Enterococcus*); TMP-SMX, trimethoprim-sulfamethoxazole; AG, aminoglycoside (**aztreonam substitutes in renal insufficiency**); MDR, multidrug-resistant; ID, infectious diseases; AUS, artificial urinary sphincter; IPP, inflatable penile prosthesis.
- **Footnotes:** a full culture-directed course for documented infection is **treatment, not prophylaxis**; penicillin allergy → clindamycin, or AG + metronidazole/clindamycin; colorectal surgery adds **oral neomycin + erythromycin or metronidazole**; **routine vancomycin is not recommended** — it is less effective against methicillin-sensitive *S. aureus*.
- **Pattern: single dose is the rule**; only PCNL and prosthetics extend to ≤24 h. Cefazolin dominates; anaerobic cover is added only where bowel or rectum is breached.
- **Traps:** simple cystoscopy with sterile urine needs **no** prophylaxis; penile surgery "likely not required"; SWL only with risk factors.

Concept — *Prophylaxis is justified by host susceptibility × inoculum × consequence, and succeeds only if a tissue level is present at the moment of contamination — hence the obsession with timing, the single dose, and redosing for long cases.*

VENOUS THROMBOEMBOLIC PROPHYLAXIS

- **Postoperative VTE is the second commonest cause** of excess stay, charges and mortality among surgical discharges.
- **Urology incidence without prophylaxis historically 10–40%**; now **1–5%** with mechanical prophylaxis and early ambulation; in the prospective @RISTOS study VTE followed open urological surgery despite high prophylaxis rates; radical cystectomy carries the highest incidence.
- **VTE is the most important cause of non-surgical mortality in urology.**
- **Mechanical prophylaxis is near-universal; pharmacological only after weighing bleeding** (Table 11.6). In general surgery, pharmacological prophylaxis increased **minor** bleeding (bruising, haematoma) but **not major** bleeding; TURP and partial nephrectomy bleed more.
- **Surgical factors:** general vs neuraxial anaesthesia; supine vs lithotomy; abdominal vs pelvic ± lymphadenectomy; open vs laparoscopic. **Patient factors:** Box 11.5.
- **ACCP 2008:** hospitals need a formal VTE strategy; moved from individual risk models to **routine group-specific prophylaxis** for each surgical group. **AUA (2009)** combines individual risk with procedure type — high-risk patient/low-risk surgery or low-risk patient/high-risk surgery may each warrant drugs (Table 11.7).

BOX 11.5 PATIENT-RELATED FACTORS INCREASING RISK FOR VENOUS THROMBOEMBOLISM

- Surgery; trauma; immobility/paresis; **cancer** (active or occult) and **cancer therapy** (hormonal, chemotherapy, angiogenesis inhibitors, radiotherapy); venous compression; **previous VTE**; **increasing age**; pregnancy/postpartum; oestrogen contraceptives or hormone replacement therapy; selective oestrogen-

receptor modulators; erythropoiesis-stimulating agents; acute medical illness; inflammatory bowel disease; nephrotic syndrome; myeloproliferative disorders; paroxysmal nocturnal haemoglobinuria; obesity; central venous catheter; thrombophilia. **Age, cancer and cancer therapy are the commonest in urology.**

TABLE 11.6 MECHANICAL AND PHARMACOLOGIC VENOUS THROMBOEMBOLISM PROPHYLAXIS

Agent	Dose	Advantages	Disadvantages
Pneumatic compression	—	Usable with high bleeding risk ; easily standardised; widely studied	No standard size/pressure; models unstudied; less effective than drugs in high risk
LMWH	40 mg SC once daily	Once daily; less heparin-induced thrombocytopenia (HIT) ; no monitoring	Not reversible ; costly; relative contraindication in renal insufficiency
Low-dose UFH	5000 U SC 8-hourly	Reversible ; safe in renal insufficiency ; cheap	Repeat 8–12-hourly; HIT

TABLE 11.7 PATIENT RISK ASSESSMENT MODEL AND AMERICAN UROLOGICAL ASSOCIATION BEST-PRACTICE RECOMMENDATIONS (MODIFIED FROM FORREST ET AL., 2009; GEERTS ET AL., 2008)

Risk	Definition	Recommendation
Low	Minor surgery, <40 years, no risk factors	Early ambulation only
Moderate	Minor surgery with risk factors; any surgery aged 40–60 without risk factors	UFH 5000 U SC 12-hourly from after surgery, or enoxaparin 40 mg SC daily, or pneumatic compression if bleeding risk high
High	Surgery >60 years; 40–60 with risk factors	UFH 5000 U SC 8-hourly, or enoxaparin 40 mg daily, or pneumatic compression if bleeding risk high
Highest	Multiple risk factors (e.g. >40, cancer, prior VTE)	Drug + pneumatic compression : enoxaparin 40 mg daily or UFH 5000 U 8-hourly

- **Enoxaparin 30 mg if creatinine clearance <30 mL/min.**
- **Dosing differs between tables:** Table 11.6 gives the standard UFH regimen (8-hourly); Table 11.7 escalates UFH from 12-hourly (moderate) to 8-hourly (high/highest).
- **Pattern:** age bands <40 / 40–60 / >60 drive stratification; only the highest tier combines drug **and** mechanical. **Trap:** compression alone is the choice only when **bleeding risk is high**.

Concept — VTE prophylaxis is a bleeding-versus-clotting trade-off: mechanical methods carry no bleeding cost but less efficacy, heparins the reverse, and the AUA model escalates from ambulation to combined therapy as patient and procedural risk accumulate.

ANAESTHETIC CONSIDERATIONS

- **Anaesthesia delivers hypnosis, amnesia and analgesia** with good operating conditions; most urology is done under monitored anaesthesia care, regional or general anaesthesia, combining inhalational and IV agents, analgesics and benzodiazepines.
- **Risk is now very low:** ambulatory all-cause mortality of the order of 1 in 50,000.

Selection of Mode of Anaesthesia

- **Urologist helps choose the mode:** patient factors (comorbidity, airway, preference) and procedural factors (complexity, duration, site, expected fluid and blood loss).

Monitored Anaesthesia Care

- **Conscious sedation under an anaesthetist**, ranging from minimal sedation to brief general anaesthesia; usually IV opioid + benzodiazepine.
- **Suits** short endoscopy, transrectal ultrasound procedures, and (with local) superficial genital surgery.
- **Office sedation**: The Joint Commission requires equivalent monitoring — trained monitoring assistant, immediate airway/resuscitation equipment, pre- and post-procedure evaluation.

Regional Anaesthesia

- **Local, spinal, epidural**. Local: avoid intravascular injection; **lidocaine vs bupivacaine differ in onset and duration**; pre-incision infiltration reduces nociceptor sensitisation and postoperative pain.
- **Spinal/epidural** (lidocaine or bupivacaine) → sensory, motor and sympathetic block.

	Spinal	Epidural
Main urological use	Endoscopy and lower abdominal surgery ; limited only by duration	Postoperative analgesia after major abdominal surgery (saves IV opioid effects)
Level and duration	Larger volume/dose → longer and more cephalad; adding low-dose opioid or vasoconstrictor prolongs analgesia at lower dose	
Complications	Hypotension (sympathetic block) 10–40% ; post-dural puncture headache <2% with 29-gauge pencil-point needles; serious neurological deficit ~0.05%	—

General Anaesthesia

Inhalational General Anaesthesia

- **Two key properties: blood/gas partition coefficient (B/G)** — uptake into blood; **minimum alveolar concentration (MAC)** — potency: the alveolar concentration preventing movement to skin incision in 50%.

COMPARISON: INHALATIONAL AGENTS

Agent	Advantages / use	Signature problem
Nitrous oxide (N₂O)	Rapid induction and emergence; low potency → combined with others	Expands closed gas spaces → contraindicated in bowel obstruction and pneumothorax ; bowel distension in laparoscopy
Halothane (1950s)	High potency	Precipitates failure with left ventricular dysfunction; sensitises myocardium to catecholamines (local adrenaline); fulminant hepatitis ~1 in 35,000
Isoflurane	Cheap (generic); low cardiac depression and sensitisation; minimal metabolism	Variable tachycardia → myocardial O ₂ demand; pungent — not for induction
Sevoflurane	Odourless → inhalational induction ; rapid; less nausea; preferred for difficult airway and severe bronchospasm	Minimal cardiac toxicity
Desflurane	Faster recovery than isoflurane for cases >3 h	Pungent — not for induction; potentiates neuromuscular blockers

Intravenous General Anaesthesia

- **Induction agent + opioid + neuromuscular blocker**; IV induction is rapid and comfortable, often with inhalational maintenance.

COMPARISON: INTRAVENOUS AGENTS

Agent	Advantage / niche	Main drawback
Thiopental	Oldest, cheapest; uncomplicated cases	Vasodilatation, cardiac depression, bronchospasm
Ketamine	Brief superficial cases; profound amnesia and somatic analgesia; raises arteriolar and bronchomotor tone → hypovolaemic and asthmatic induction	—
Propofol	Commonest (outpatients); rapid; bronchodilator; smooth nausea-free emergence	Hypotension
Midazolam	Profound amnesia and anxiolysis; rapid, short, little cardiac effect	Never used alone
Fentanyl (synthetic opioid)	Most widely used opioid: ~50–100× morphine potency , rapid onset, short action; opioids spare other agents	—

- **Neuromuscular blockade** gives full relaxation without toxic doses of other agents.

	Depolarising	Non-depolarising
Mechanism	Depolarises the muscle membrane, making it refractory to acetylcholine	Blocks acetylcholine binding at pre- and postsynaptic receptors
Agent	Succinylcholine — the only one ; rapid onset (rapid-sequence induction), ~5 min duration	Several; differ in metabolism; potentiated by drugs including desflurane
Risks	Malignant hyperthermia (with volatiles), hyperkalaemia , bradycardia in children	Used when succinylcholine contraindicated

- **Commonest complication: inadequate reversal** → respiratory failure, reintubation, pulmonary complications.
- **Train-of-four (TOF)** ratio: fourth twitch ÷ first, ulnar stimulation at 0.5-s intervals. **Historical threshold 0.7** → **now 0.9** for full recovery.
- **Clinical tests** (head lift, tongue depressor, grip) miss residual block in many patients → **quantitative TOF (acceleromyography) plus clinical tests before extubation**.

Concept — Each anaesthetic agent is chosen for its signature advantage and avoided for its signature toxicity: know one niche and one danger per drug, and the choice for a given patient follows.

SKIN PREPARATION

- **Antiseptics**: alcohol, povidone-iodine or chlorhexidine; apply in **concentric circles from the centre** and **let dry** before incision.
- **Cochrane**: some evidence that **0.5% chlorhexidine in methylated spirits** beats alcohol-based povidone-iodine for clean surgery; no conclusive superiority overall.
- **CDC recommends preoperative showering**, but antiseptic bathing does not reduce infection.
- **Hair removal, if done: immediately before surgery, with clippers** (shaving breaches skin → colonisation).

TRANSFUSION CONSIDERATIONS

- **Urological organs are vascular**; know transfusion risks and alternatives (e.g. Jehovah's Witnesses).
- **Before the acquired immunodeficiency syndrome (AIDS) epidemic**: transfusion for haematocrit <30%; infection fears → National Institutes of Health consensus (1988) → **ASA guidelines (2006)**.

Haematocrit	Transfusion
>30%	Rarely indicated
21–30%	Individualise — oxygenation risk; moderate–severe comorbidity (lung disease, coronary disease, vascular insufficiency, shock) may not tolerate <30%
<21%	Often indicated; minor comorbidity tolerates >21%

- **Component therapy:** packed red cells (PRBCs) = whole blood minus plasma; **haematocrit 40% whole blood vs ~70% PRBC**; given with crystalloid.
- **Platelets:** not empirical; give if **<50,000/mm³ with planned surgery**, or **50,000–100,000 with high-risk surgery or platelet dysfunction**.
- **Fresh frozen plasma (FFP):** **urgent warfarin reversal; specific factor deficiency; bleeding with INR >1.5**; in massive transfusion without an INR, **after one blood volume**. Not empirical.

COMPARISON: TRANSFUSION HAZARDS

Hazard	Key features
Haemolytic reaction (ABO or non-ABO)	Fever, chills, chest pain, hypotension, bleeding during/just after; may be delayed (recipient antibodies). Stop transfusion, fluids, alkalinise urine
TRALI (transfusion-related acute lung injury)	Commonest cause of transfusion death (55% in 2005–07); non-cardiogenic pulmonary oedema 1–2 h after; supportive; most recover
Viral	Nucleic-acid testing → human immunodeficiency virus and hepatitis C ~1 in 2 million; hepatitis B ~1 in 200,000
Bacterial	Highest with platelets ~1 in 5000 units

- **Massive haemorrhage:** components traditionally after >6 units PRBC; trauma data favour higher platelet:FFP:red-cell ratios — **massive transfusion protocol triggered by anticipated >10 units in 24 h**.
- **Jehovah's Witnesses:** refuse the four major components (RBCs, white cells, platelets, plasma), **not medical care**; some accept albumin, immunoglobulins, clotting factors — agree in advance with the anaesthetist. Options: diathermy, **cell salvage, intraoperative haemodilution**, high-dose erythropoietin, iron, vitamins C and D. Haemoglobin-based substitutes are not yet usable.

PAIN MANAGEMENT

- **Untreated pain adds physiological stress** and complications; analgesia blocks nociception somewhere between receptor, afferent nerve, cord and cortex.

COMPARISON: POSTOPERATIVE ANALGESIC CLASSES

Class	Action / benefit	Cautions
Opioids	Central modulation (dorsal root ganglion to cortex); oral, IV, neuraxial or transdermal by severity and oral tolerance	Ileus, respiratory depression, sedation, confusion; weaker combinations with paracetamol (hydrocodone, codeine) — caution in liver disease; addiction and the opioid epidemic
NSAIDs	Cyclo-oxygenase inhibition → less prostaglandin (the tissue-level nociceptor mediator); opioid-sparing, less nausea ; IV ketorolac improved pain and ileus after laparoscopic colectomy	Avoid in renal insufficiency ; caution with reflux or peptic ulcer

Class	Action / benefit	Cautions
Epidural	Opioid + local anaesthetic, continuous and/or patient-controlled; better pain control than parenteral opioids (meta-analysis); with general anaesthesia reduces perioperative mortality and multi-organ morbidity	Run by anaesthesia/pain team

- **Overprescribing:** most prescribed opioids go unused and seed misuse; **new persistent opioid use follows ~6% of minor and major operations.**
- **Prescribing varies widely** after stone treatment; after ureteroscopy, patients managed without opioids sought care for pain no more often — **ureteroscopy can be managed opioid-free.**

Concept — *Modern perioperative analgesia is opioid-sparing: NSAIDs and regional techniques deliver equal or better pain control while avoiding the ileus, respiratory depression and community harm that opioids carry.*

KEY POINTS

- **Proper preoperative evaluation** prevents cancellations and reduces complications.
- **Cardiac testing depends on** functional capacity, cardiac risk factors and surgery-specific risk.
- **SSI** is a leading complication; prophylaxis is indicated in virtually all procedures.
- **VTE:** AUA recommends mechanical or pharmacological prophylaxis (both for the highest risk).
- **Anaesthetic pharmacology** lets the urologist help choose the mode.
- **Transfuse at haematocrit <21%** in most, unless cardiopulmonary risk.
- **Hypothermia of even 1–2°C** increases blood loss and SSI.